

NYCU Introduction to Machine Learning, Homework 4

[112550018], [吳欣澤]

Part. 1, Kaggle (70% [50% comes from the competition]):

(20%) Introduction of your idea, methods, and key to success

Create a 3-5 page slide (e.g., MS PowerPoint, Google Slides, etc.) with no title or thank-you page to introduce your work.

Things you should cover - Include and not limit to

- How do you process the data? Have you done any special processing that significantly boosted performance?
- What is your model architecture? Do you make any changes/modifications to the model? Does it improve performance?
- How do you train the model? Do you use any special techniques (e.g., ensembles or other methods) to improve performance?
- Other details you want to mention that improve the performance
- Paste the snapshot of your Kaggle public leaderboard as an appendix (Not count to the page limit)

Hint:

1. Make your slide presentation clear and informative, and TAs will evaluate its completeness and content.
2. Read some literature to see how they report their method and results.

Part. 2, Questions (30%):

1. (10%) Based on the “SVMs vs. Logistic regression” lecture slide, explain which kinds of training data points mainly determine the classifier learned by SVM and which types of points influence Logistic Regression, and briefly justify your answer by referring to the shapes of hinge loss and logistic loss.

Only the support vectors mainly determine the classifier learned by SVM. For the logistic regression, all training data. Referring to the shapes of hinge loss and logistic loss, we can see that the hinge loss = 0 if $z \geq 1$, which means the gradient is also zero, so only for those incorrectly classified points or points in the margin ($z < 1$) would have loss and so the gradient, but for the logistic loss was always > 0 over all z , so all data points would influence.

2. (15%) For an SVM with a Linear Kernel, determine whether to use the Primal or Dual Form for the datasets below. Justify your choice based on: (a) Optimization variables & computational complexity, (b) Memory requirements (specifically the size of the Gram Matrix), (c) Prediction cost.

- Dataset A: $N=100$, $M=20,000$
- Dataset B: $N=1,000,000$, $M=20$

A: Dual

Optimization (A): The Dual form optimizes 100 variables (α 's). The Primal form would optimize 20,000 variables (w 's). Fewer variables means much faster training.

Memory (B): The Dual form's critical piece of memory is the Gram Matrix, which is $N \times N$. This is tiny and easy to manage.

Prediction (C): Prediction is fast, as it depends on a small number of samples (the Support Vectors ≤ 100).

B: Primal

Optimization (A): The Primal form optimizes only 20 variables (w 's). The Dual form would try to optimize 1,000,000 variables, which would be very slow.

Memory (B): The Dual form requires a Gram Matrix of $N \times N$, or $1,000,000 \times 1,000,000$. This is millions of gigabytes and impossible to store in memory. The Primal form avoids this memory issue.

Prediction (C): Prediction is extremely fast. It's just a dot product in the 20-dimensional space, which is highly efficient.

3. (5%) To train a neural network, what do we need to optimize it? (How do we know the network is good or not?) Also, what algorithm can we use to optimize the neural network? (the most basic one).

To train a neural network, you need to optimize the weights and biases across all layers of the network. We know how "good" the network is by calculating the output of the Error Function (or Loss Function); a lower value means better performance. The basic and fundamental algorithm used to find the optimal weights and biases that minimize this loss is the combination of Gradient Descent and Backpropagation, where Backpropagation efficiently calculates the necessary gradients, and Gradient Descent uses those gradients to iteratively adjust the parameters.